STATISTICS & MACHINE LEARNING · LESSON 4 OF 28

Hypothesis Testing I: The Framework, p-values, z-tests and t-tests

Turn business questions into testable hypotheses. Null and alternative hypotheses, significance level, test statistics, p-values and critical values, one- vs two-tailed tests, Type I and II errors and power, then one-sample, two-sample (Welch) and paired t-tests and z-tests for proportions, on a delivery-time case study.

ML classes 1–2 · SciPy 1.1x · statsmodels 0.1x · Intermediate · 120 min

By the end of this lesson you can

- Translate a business question into null and alternative hypotheses
- Explain significance level, test statistic, p-value and critical value, and decide with both
- Choose one- or two-tailed tests and describe Type I/II errors and power
- Run and interpret one-sample, independent (Welch) and paired t-tests and a z-test for proportions
- Report effect sizes and confidence intervals, not just p-values

Datasets used

`delivery_times.csv` `hr_employees.csv`

From business question to hypothesis

> **Case.** A quick-commerce company promises "delivery in 30 minutes". Operations claim Pune meets the promise on average. Is the average delivery time in Pune really 30 minutes, or is it slower? Is Hyderabad different from Pune?

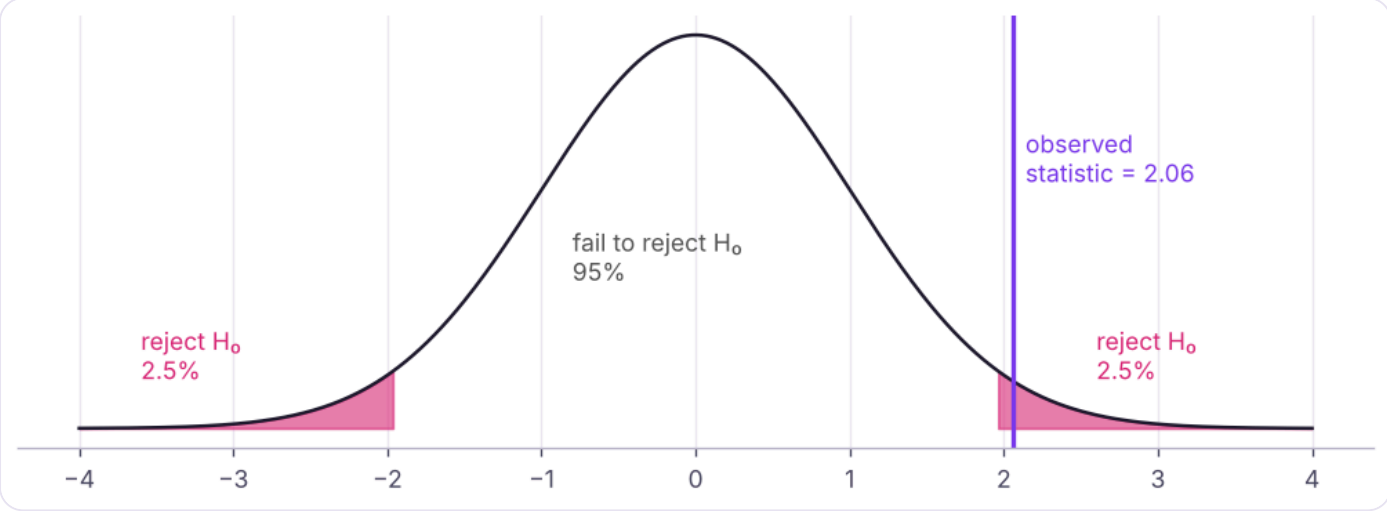
A **hypothesis test** checks whether data are consistent with a default claim.

Term	Meaning	Delivery case
H_0 (null)	No effect / status quo; assumed true until evidence	μ_Pune = 30 min
H_1 (alternative)	What we want evidence for	μ_Pune > 30 min (one-tailed) or ≠ 30 (two-tailed)
α (significance level)	Accepted risk of a false alarm	0.05
Test statistic	Distance from H_0 in standard-error units	$t = (\bar{x} - 30)/(s/\sqrt{n})$
p-value	P(data at least this extreme | H_0 true)	from the t distribution
Decision	$p \le \alpha$ → reject H_0; $p > \alpha$ → fail to reject H_0	



The hypothesis-testing procedure.

p-values and critical values



Two-tailed test at α = 0.05: the rejection regions are the 2.5% tails beyond ±1.96 (z). The p-value is the tail area beyond the observed statistic.

Python

```
print("critical values: two-tailed z ±", round(stats.norm.ppf(0.975), 3), "| one-tailed z",
round(stats.norm.ppf(0.95), 3), "| two-tailed t(df=59) ±", round(stats.t.ppf(0.975, 59),
3))
for z in (1.2, 1.7, 2.06, 3.1): print(f"z = {z}: two-tailed p = {2 *
stats.norm.sf(abs(z)):.4f} | one-tailed p = {stats.norm.sf(z):.4f}")
```

OUTPUT

```
critical values: two-tailed z ± 1.96 | one-tailed z 1.645 | two-tailed t(df=59) ±
2.001
z = 1.2: two-tailed p = 0.2301 | one-tailed p = 0.1151
z = 1.7: two-tailed p = 0.0891 | one-tailed p = 0.0446
z = 2.06: two-tailed p = 0.0394 | one-tailed p = 0.0197
z = 3.1: two-tailed p = 0.0019 | one-tailed p = 0.0010
```

WATCH OUT **What a p-value is not**

It is **not** the probability that H_0 is true, and 1 − p is not the probability that H_1 is true. A large p-value does not prove H_0 ("absence of evidence is not evidence of absence"). A tiny p-value does not mean a large or important effect.

One- vs two-tailed

	Two-tailed	One-tailed
H_1	$\mu \neq 30$ (any difference)	$\mu > 30$ (only "slower")
α split	2.5% in each tail	5% in one tail
When	Default; direction unknown	Direction fixed **before** looking at data, and the other direction is irrelevant

Errors and power

	H_0 true	H_0 false
Reject H_0	**Type I error** (false positive), probability α	Correct: **power** = 1 − β
Fail to reject	Correct	**Type II error** (false negative), probability β

Fire alarm analogy: a Type I error rings without a fire; a Type II error stays silent during a fire. Lower α means fewer false alarms but more misses unless you collect more data. Power (commonly 80%) grows with sample size, effect size and α.

One-sample t-test: does Pune meet 30 minutes?

Python

```
import pandas as pd
d = pd.read_csv("delivery_times.csv")
pune, hyd = d[d.city == "Pune"].delivery_minutes, d[d.city == "Hyderabad"].delivery_minutes
print(f"Pune: n={len(pune)}, mean {pune.mean():.2f}, sd {pune.std():.2f}")
t_two = stats.ttest_1samp(pune, 30)
t_one = stats.ttest_1samp(pune, 30, alternative="greater")
se = pune.std() / np.sqrt(len(pune))
print(f"t = {t_two.statistic:.3f}, df = {len(pune) - 1} | two-tailed p = {t_two.pvalue:.4f}
| one-tailed (greater) p = {t_one.pvalue:.4f}")
print(f"by hand: t = ({pune.mean():.2f} - 30) / {se:.3f} = {(pune.mean() - 30) / se:.3f}")
ci = stats.t.interval(0.95, len(pune) - 1, loc=pune.mean(), scale=se)
print(f"95% CI for mean: {ci[0]:.2f} to {ci[1]:.2f} minutes (30 is {'outside' if not ci[0]
⩽ 30 ⩽ ci[1] else 'inside'})")
print(f"effect size (Cohen's d) = {(pune.mean() - 30) / pune.std():.2f}")
```

OUTPUT

```
Pune: n=60, mean 31.70, sd 6.39
t = 2.060, df = 59 | two-tailed p = 0.0439 | one-tailed (greater) p = 0.0219
by hand: t = (31.70 - 30) / 0.825 = 2.060
95% CI for mean: 30.05 to 33.35 minutes (30 is outside)
effect size (Cohen's d) = 0.27
```

Conclusion. At α = 0.05 we reject H_0: Pune's average delivery time (about 31.7 minutes) is statistically higher than the 30-minute promise (p ≈ 0.02 one-tailed). The difference is small (under 2 minutes, d ≈ 0.27), so the business question is whether 1–2 minutes matter to customers, not just whether p < 0.05.

Two-sample t-test: Pune vs Hyderabad

Python

```
print(f"Hyderabad: n={len(hyd)}, mean {hyd.mean():.2f}, sd {hyd.std():.2f}")
lev = stats.levene(pune, hyd); print(f"Levene test for equal variances p = {lev.pvalue:.3f}
→ {'unequal' if lev.pvalue < 0.05 else 'equal'} variances")
w = stats.ttest_ind(hyd, pune, equal_var=False)            # Welch's t-test: safe default
print(f"Welch t = {w.statistic:.3f}, p = {w.pvalue:.4f}, difference = {hyd.mean() -
pune.mean():.2f} min")
pooled = np.sqrt(((len(pune) - 1) * pune.var() + (len(hyd) - 1) * hyd.var()) / (len(pune) +
len(hyd) - 2))
print(f"Cohen's d = {(hyd.mean() - pune.mean()) / pooled:.2f}")
```

OUTPUT

```
Hyderabad: n=55, mean 33.81, sd 7.60
Levene test for equal variances p = 0.047 → unequal variances
Welch t = 1.601, p = 0.1123, difference = 2.11 min
Cohen's d = 0.30
```

Conclusion. p ≈ 0.11 > 0.05: we **fail to reject** H_0. The data do not show a significant difference between the cities (Hyderabad is about 2 minutes slower in this sample, but that could be chance). Use **Welch's test** (`equal_var=False`) by default: it does not assume equal variances and loses almost nothing when they are equal.

Paired t-test: before and after

Use a **paired** test when the same units are measured twice (same riders before/after training, same stores before/after a layout change).

Python

```
rng = np.random.default_rng(5)
before = rng.normal(34, 5, 30).round(1)
after = (before - rng.normal(1.5, 1.5, 30)).round(1)          # simulated route-training
effect
pt = stats.ttest_rel(before, after)
it = stats.ttest_ind(before, after, equal_var=False)
print(f"mean improvement {np.mean(before - after):.2f} min | paired p = {pt.pvalue:.4f} |
(wrong) independent p = {it.pvalue:.4f}")
```

OUTPUT

```
mean improvement 1.17 min | paired p = 0.0003 | (wrong) independent p = 0.3342
```

Pairing removes rider-to-rider variation, so the same improvement is detected far more clearly.

z-test for proportions

Python

```
from statsmodels.stats.proportion import proportions_ztest
hr = pd.read_csv("hr_employees.csv")
ot = hr[hr.overtime == "Yes"].attrition.eq("Yes"); no = hr[hr.overtime ==
"No"].attrition.eq("Yes")
stat, p = proportions_ztest([ot.sum(), no.sum()], [len(ot), len(no)])
print(f"attrition: overtime {ot.mean():.1%} (n={len(ot)}) vs no overtime {no.mean():.1%}
(n={len(no)}) | z = {stat:.2f}, p = {p:.2e}")
stat1, p1 = proportions_ztest(hr.attrition.eq("Yes").sum(), len(hr), value=0.20,
alternative="larger")
print(f"is company attrition above the 20% industry benchmark? z = {stat1:.2f}, p =
{p1:.2e}")
```

OUTPUT

```
attrition: overtime 36.7% (n=433) vs no overtime 21.6% (n=1067) | z = 6.07, p =
1.26e-09
is company attrition above the 20% industry benchmark? z = 5.24, p = 7.89e-08
```

Choosing a test (means)

Situation	Test	Python
One mean vs a target, σ unknown	One-sample t	`stats.ttest_1samp`
Two independent groups	Welch's t	`stats.ttest_ind(..., equal_var=False)`
Same units twice	Paired t	`stats.ttest_rel`
One or two proportions	z-test	`statsmodels proportions_ztest`
Non-normal small samples	Mann-Whitney U / Wilcoxon (next lesson)	`stats.mannwhitneyu`, `stats.wilcoxon`
3+ groups	ANOVA (next lesson)	`stats.f_oneway`

Assumptions for t-tests: independent observations, roughly normal data **or** large samples (CLT), and for the pooled version equal variances.

Reporting

> Pune deliveries averaged **31.7 minutes** (95% CI 30.1–33.4, n = 60), significantly above the 30-minute promise (one-sample t = 2.06, one-tailed p = 0.022). The gap is small (d = 0.27). **Recommendation:** monitor the share of orders above 40 minutes (13%) rather than the average.

Report the **estimate, CI, test, p-value, effect size and business meaning**.

Summary

KEY TAKEAWAYS

- H_0 is the default (no effect); H_1 is what you want evidence for; fix α (usually 0.05) and the tails before looking.
- p-value = probability of data this extreme if H_0 were true; reject H_0 when $p \le \alpha$ (equivalently when the statistic passes the critical value).
- Type I (false positive, α) vs Type II (false negative, β); power = 1 − β grows with n and effect size.
- One-sample, Welch two-sample and paired t-tests compare means; z-tests compare proportions.
- Always report effect size and a confidence interval: statistical significance is not business significance.

Quick quiz

6 questions · tap an answer to check it

1. The p-value is 0.03 and $\alpha = 0.05$. What do you conclude?

- **A** Reject H_0: the result is statistically significant
- **B** Accept H_1 with 97% probability
- **C** Fail to reject H_0
- **D** H_0 is true

Answer: A. $p \leq \alpha$.

2. Which is a Type I error?

- **A** Concluding a new checkout works when it actually has no effect
- **B** Missing a real improvement
- **C** Using the wrong chart
- **D** A small sample

Answer: A. A false positive.

3. Which test compares the same riders before and after training?

- **A** Welch's t-test
- **B** Paired t-test
- **C** Chi-square test
- **D** One-sample z-test

Answer: B. Measurements are paired by rider.

4. Why prefer Welch's t-test for two groups?

A It does not assume equal variances and works well either way

B It is always more significant

C It needs no data

D It is for proportions

Answer: A. `equal_var=False`.

5. p = 0.40. What can you say?

A H_o is proven true

B The data do not provide enough evidence against H_o

C The effect is 40%

D The test failed

Answer: B. Absence of evidence is not evidence of absence.

6. What increases the power of a test?

A Smaller sample

B Larger sample size or larger true effect

C Lower α only

D Using a two-tailed test always

Answer: B. Power = probability of detecting a real effect.

EXERCISE

Practice

(1) Test whether Hyderabad's mean delivery time exceeds 30 minutes. (2) Test whether employees who left had lower monthly income than those who stayed (one-tailed Welch). Report the difference and d. (3) Test whether the share of overtime workers differs between Sales and IT.

Show solution (worked answer)

Python

```
print(stats.ttest_1samp(hyd, 30, alternative="greater"))
left, stay = hr[hr.attrition == "Yes"].monthly_income, hr[hr.attrition ==
"No"].monthly_income
r = stats.ttest_ind(left, stay, equal_var=False, alternative="less"); print(r,
left.mean() - stay.mean())
print((left.mean() - stay.mean()) / np.sqrt((left.var() + stay.var()) / 2))
s_, i_ = hr[hr.department == "Sales"].overtime.eq("Yes"), hr[hr.department ==
"IT"].overtime.eq("Yes")
print(proportions_ztest([s_.sum(), i_.sum()], [len(s_), len(i_)]))
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain a p-value to a business stakeholder. If there were truly no difference, how surprising would our data be? A p-value of 0.01 means a result this extreme would happen only 1% of the time by chance alone.

Q: What is the difference between statistical and practical significance? Statistical significance says an effect is unlikely to be zero; practical significance asks whether it is large enough to matter (effect size, cost, revenue).

Q: How do you choose α? By the cost of errors: 0.05 by convention; lower (0.01) when false positives are expensive (medical, financial launches); sometimes higher for early exploration.

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